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RESEARCH ARTICLE



High-pressure study on novel structure, mechanical properties and high energy density of RuN₄

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ABSTRACT

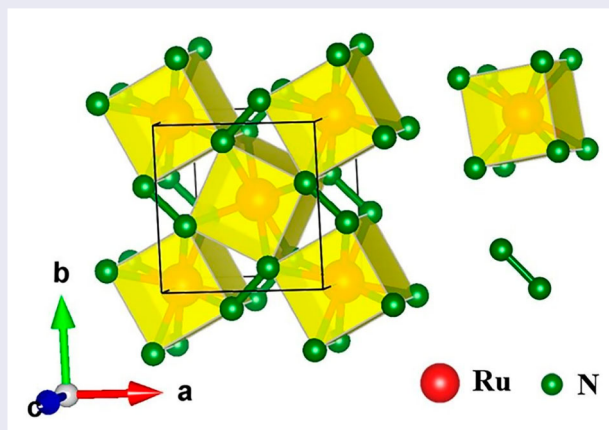
Novel structures with high energy density could be created with the addition of transition-metal elements into polymeric nitrogen. Based on *ab initio* evolutionary algorithm for crystal structure prediction and first-principles calculations, we have conducted a comprehensive search on the crystal structure of RuN₄ under the pressure range of 0–300 GPa and uncovered its pressure-induced phase transition. The consequences indicate that RuN₄ decomposes into Ru and N₂ under ambient pressure. Subsequently, RuN₄ can stabilise with the triclinic $P\bar{1}$ -RuN₄ phase at 37 GPa, which transforms into the tetragonal $P4/mbm$ -RuN₄ structure at 62 GPa. Interestingly, the $P\bar{1}$ -RuN₄ phase contains RuN₆ distorted octahedron and extended chain (N_∞), while the $P4/mbm$ -RuN₄ structure is composed of RuN₈ rectangles and N₂ units. Mechanical property analyses have revealed that the $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ crystals possess high hardness, excellent compression resistance and strong rigidity. Intriguingly, the Vickers hardness value for the $P4/mbm$ RuN₄ phase can reach 33.50 GPa. In addition, the electronic structure calculations demonstrate both phases are metallic. Moreover, strong Ru–N and N–N covalent bonds exist in both structures. At ambient conditions, the $P4/mbm$ RuN₄ phase has an energy density of 2.81 kJ/g and a volumetric energy density of 18.70 kJ/cm³, which is considered as an excellent high-energy density material. Hence, the $P4/mbm$ RuN₄ crystal is a newly multifunctional material that possesses both high hardness and high-energy density.

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The $P4/mbm$ -RuN₄ consists of RuN₈ rectangles and N₂ polymers. In the RuN₈ rectangular unit, the ruthenium atoms occupy the centre of the tetragonal lattice and the upper and lower lateral prisms, and each ruthenium atom is surrounded by eight equally spaced nitrogen atoms.

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1. Introduction

Nitrogen is a highly abundant element in the atmosphere and belongs to Group VA of the periodic table. Moreover, nitrogen atoms have five valence electrons ($2s^2 2p^3$) with extremely complex bonding modes [1]. Under environmental conditions, Nitrogen atoms in nitrogen gas exist in the type of $N \equiv N$ triple bonds while nitrogen molecules within the covalent triple bonds dissociate and promote the formation of polymeric nitrogen structure under high pressure [2–5]. Polymeric nitrogen is a candidate for green high-energy-density materials (HEDMs) because of the significant amount of energy required for the decomposition of $N-N$ single or $N=N$ double bonds polymeric nitrogen into $N \equiv N$ triple bonds nitrogen dimers. These single (~ 167 kJ/mol) and double (~ 419 kJ/mol) bonds have relatively low bond energies compared with the $N \equiv N$ bonds (~ 954 kJ/mol). In addition, experimenting with extreme conditions can result in the formation of single and double nitrogen bonds (e.g. high temperature and pressure) [6–10]. In recent years, researchers have probed into the structure and properties of polymeric nitrogen and have achieved fruitful results. For example, Mailhot *et al.* [8, 11] and Wang *et al.* [12] theoretically reported polymeric nitrogen networks like cg-N structures and layered PP structures. Additionally, Eremets *et al.* [6] and Laniel *et al.* [13] successfully synthesised cg-N (110 GPa, 2000K) and HLP-N (244 GPa, 3300 K) polymeric nitrogen structures by laser-heated diamond top anvil experimental technique. It is found that the synthesis conditions are extremely rigorous, which restricts the application of polymeric nitrogen and stimulates us to explore more novel polymeric nitrogen structures.

Recent results show that the introduction of coordination elements M into polymeric nitrogen can lead the generation of novel polymeric nitrogen structures due to excellent advantages such as unique nitrogen linkage, mild synthesis conditions, high stability and high-energy density. For example, alkali metal azides NaN_3 [14], KN_3 [15] and CsN_3 [16] have N_6 rings polymeric nitrogen structures. Furthermore, LiN_5 [17] is synthesised at 45 GPa and can be quenched to ambient conditions. Alkaline earth metal polyazides MgN_4 [18] and CaN_4 [19] have armchair chain polymeric nitrogen structures and $(CaN_6)_n$ [20] has a linear polymeric nitrogen structure. Moreover, $P21/c$ - BeN_4 [21] has an energy-stable state of around 40 GPa and can be quenched and recovered to ambient conditions, made up of a unique polynitrogen and extended polymeric 3D puckered N_{10} rings network. At the same time, many polymeric nitrogen compounds have high energy density properties, such as

BeN_4 (6.35 kJ/g) [21], BeN_{10} (5.39 kJ/g) [22] and AlN_{15} (5.31 kJ/g) [23].

The addition of transition metal elements with complex electronic configurations into nitrides may produce richer polymeric nitrogen structures than the main group metal elements [24–28]. Niu *et al.* [29] reported for the first time the existence of N_{14} network structure in $C2/m$ - CdN_6 compounds. A theoretical investigation [30] indicates the nitrogen atoms are in the form of N_2 dimers in T - ReN_8 under lower pressure, whereas zigzag polynitrogen chains exist in M -, T' - and M' - ReN_8 under high pressure. Moreover, Wei *et al.* [31] displayed that a structure consists of six-membered ' N_6 ' rings that resemble benzene, and they are arranged with specific Mo cations (MoN_6) or Re cations (ReN_6). The novel N_7 rings and N_{20} rings were first found in the folded layer structure of AgN_7 [32]. Liu *et al.* [33] show that CuN_5 containing $cyclo-N_5^-$ exhibits remarkable thermodynamic stability under ambient conditions. Interestingly, it was first reported that CoN_8 comprises the N-structure that has been redesigned to have a layer shape while CoN_{10} consists of the novel N-structure with a band-shaped pattern [34]. Moreover, Nitrogen-rich HfN_{10} [35] exhibits an infinite armchair polymer N-chain structure at pressures up to 23 GPa. From the above review, we know that multi-nitrogen structures can possess unique nitrogen linkages, milder synthesis conditions, higher energy density and higher stability than pure nitrogen polymer structures. Therefore, it is desirable to probe into the new type of polynitrogen configurations by considering more transition metal elements.

Ru is a typical transition metal element and has a complex electronic structure. Among the Ru–N binary system, Ram *et al.* [36] first observed a RuN electronic transition with a Fourier transform spectrometer. Subsequently, a great deal of research has been done on the structures and properties associated with transition metal ruthenium nitrides. Moreno-Armenta *et al.* [37] used reactive pulsed laser ablation to synthesise RuN and suggested that transition metal ruthenium nitrides could be superhard materials under environmental conditions. Dong *et al.* [38] compared the elastic modulus of RuN_2 with several other isostructural noble metal nitrides and found that RuN_2 (566 GPa) is almost comparable to OsN_2 (579 GPa) and IrN_2 (572 GPa), implying that RuN_2 is a potential super-incompressible hard material. In 2016, Zhang *et al.* [39] performed a structural search for stable phases in RuN_x ($x = 1-4$) under the pressure range of 0–50 GPa, which also predicted the Vickers hardness values of 20.1–23.4 GPa. Although transition metal nitrides RuN_x have been widely studied as potential hard materials, there is a lack of research

on them as HEDMs. Transition metal nitrides with high nitrogen content tend to form polymeric nitrogen structures with high-energy density. So far, polymeric nitrogen structures have not been reported in the theoretical studies of RuN_4 compounds. Therefore, our purpose is to uncover polymeric nitrogen structures with high-energy density by investigating polynitrogen structures of RuN_4 under high pressure.

In this work, we have employed Universal Structure Predictor: Evolutionary Xtallography (USPEX) structure search in conjunction with first-principle calculations to perform structure searches for RuN_4 in the pressure range of 0–300 GPa. We found two stable structures of RuN_4 , $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 . $P\bar{1}$ - RuN_4 is stable in the range of 37–162 GPa while $P4/mbm$ - RuN_4 could be stable under the range of 162–300 GPa. Meanwhile, we have also investigated the electronic properties, mechanical properties, chemical bonding and energy densities of these two stable structures in detail. It is worth noting that $P4/mbm$ - RuN_4 is expected to release a significant amount of energy (2.81 kJ/g) upon decomposition with a volumetric energy density of 18.70 kJ/cm³. Hence, $P4/mbm$ - RuN_4 can be considered as a high-energy density material.

2. Computational methods

To obtain the stable and metastable phases of RuN_4 under ambient and high pressures, we carried out a comprehensively structural search of RuN_4 through the evolutionary algorithm, as conducted in USPEX (universal structure predictor: Evolutionary Xtallography) code [40–42]. The structure predictions were implemented in the pressure range of 0–300 GPa with up to four formula units (f.u) at zero temperature. The following structural optimisation and electronic structure calculations were conducted within the projector augmented wave (PAW) method in the Vienna *ab initio* simulation package (VASP) [43]. We employed the Perdew–Burke–Ernzerhof (PBE) [44] exchange–correlation function in generalised gradient approximation (GGA). In addition, the PAW pseudopotential was utilised for Ru and N atoms with valence electrons of $4p^6 5s^1 4d^7$ and $2s^2 2p^3$, respectively. The cut-off energy of 600 eV for the plane-wave basis expansion and appropriate Monkhorst–Pack [45] Brillouin sampling grid of spacing $2\pi \times 0.03 \text{ \AA}^{-1}$ were chosen to verify that all the enthalpy calculations are perfectly converged to lower than 1 meV/atom. Phonon dispersion was computed through the finite displacement method in the PHONOPY software package [46]. Chemical bonds and interatomic interactions were analysed based on the crystal orbital Hamilton population (COHP) method in the LOBSTER programme [47, 48]. The electron localisation

function (ELF) [49] was calculated via the VASP code. In addition, the crystal structure and ELF were plotted with the VESTA [50] software. On account of the strain–stress method and Voigt–Reuss–Hill [51] approximation, the crystal elastic constants, the bulk modulus, shear modulus, Young’s modulus, and Poisson’s ratio of the considered structures were predicted. The Vickers hardness can be estimated by using the empirical formula $H_v = 2(k^2 G)^{0.585} - 3$ [52], where the shear modulus and the Pugh’s modulus ratio are represented by G and $k = G/B$ in separate equations.

3. Results and discussion

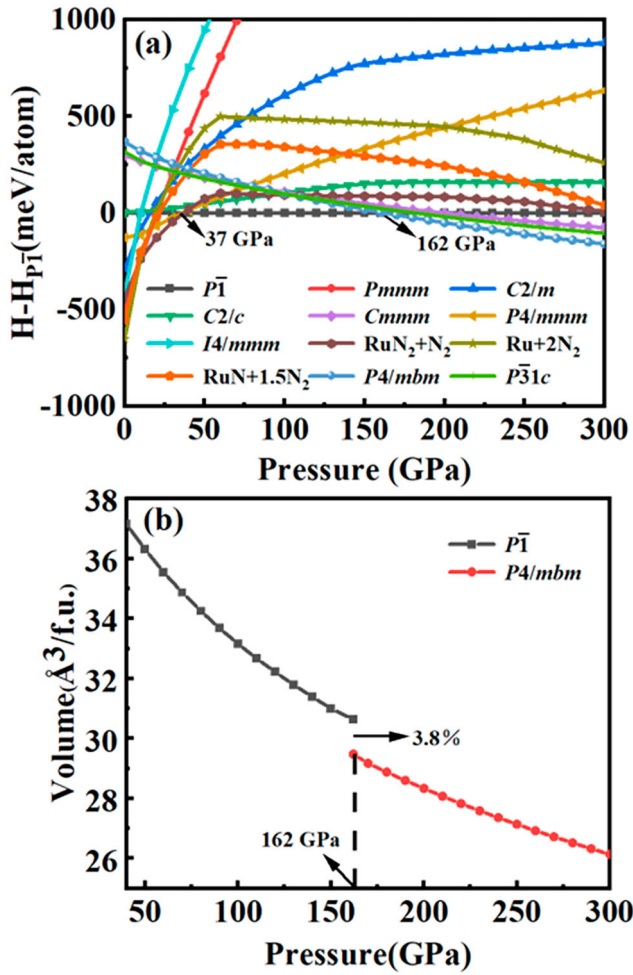
3.1. Crystal structure phase transition and structural characteristics

Pressure is one of the important physical quantities in thermodynamics and can affect the stability of a material [53–57]. To predict the structural transformation and the energetically ground state for RuN_4 under the pressure range of 0–300 GPa, we performed structure prediction of RuN_4 within the USPEX structure prediction method. Firstly, eight candidate structures of RuN_4 were identified, reproducing three previously reported metastable structures, namely, $P4/mmm$, $C2/c$ and $Cmmm$ [39], proving the reliability of the USPEX methodology used in structure prediction in RuN_4 . Subsequently, the enthalpy–pressure diagram for RuN_4 with respect to the $P\bar{1}$ phase was depicted in Figure 1(a). The RuN_4 compound is unstable at atmospheric pressure and decomposes into Ru and N_2 . When the pressure increases to 37 GPa, RuN_4 becomes stable with the triclinic $P\bar{1}$ structure. Upon further compression, a tetragonal $P4/mbm$ configuration is confirmed within the pressure range of 162–300 GPa. Therefore, RuN_4 undergoes a structural transition from $P\bar{1}$ - RuN_4 to $P4/mbm$ - RuN_4 with a transition pressure of 162 GPa. This can also be verified by the volume–pressure curve (see Figure 1(b)), where an apparent volume collapse rate of 3.8 is found at 162 GPa. Hence, we mainly consider the structural and correlative properties of the $P\bar{1}$ - RuN_4 phase at 50 GPa and the $P4/mbm$ - RuN_4 phase at 200 GPa.

To extensively uncover the structural features of the $P\bar{1}$ - RuN_4 phase at 50 GPa and the $P4/mbm$ - RuN_4 phase at 200 GPa, we summarise their structural parameters in Table 1 and construct their polyhedral structure and N–N link in Figures 2 and 3, respectively. For the $P\bar{1}$ - RuN_4 structure at 50 GPa, the lattice constants are $a = 2.740 \text{ \AA}$, $b = 3.710 \text{ \AA}$, $c = 3.933 \text{ \AA}$, and the lattice intercalation angles are $\alpha = 104.802^\circ$, $\beta = 92.532^\circ$, $\gamma = 108.528^\circ$. Moreover, the Wyckoff occupation of Ru atoms are 1 *h* (0.500, 0.500, 0.500), and the Wyckoff

Table 1. Crystal structure information of two stable RuN₄ at high pressure.

Phase	Lattice parameters (Å, deg)	Atom coordinates (fractional)			
		Atom	<i>x</i>	<i>y</i>	<i>z</i>
<i>P</i> $\bar{1}$ -RuN ₄ (50 GPa)	<i>a</i> = 2.740	N1(2 <i>i</i>)	0.859	0.998	0.858
	<i>b</i> = 3.710	N3 (2 <i>i</i>)	0.945	0.361	0.825
	<i>c</i> = 3.933	Ru1(1 <i>h</i>)	0.500	0.500	0.500
	α = 104.802				
	β = 92.532				
<i>P</i> 4/ <i>mbm</i> -RuN ₄ (200 GPa)	γ = 108.528				
	<i>a</i> = 4.080	Ru1(2 <i>b</i>)	0.000	0.000	0.500
	<i>b</i> = 4.080				
	<i>c</i> = 3.403	N1(8 <i>k</i>)	0.880	0.620	0.198
	α = 90.000				
	β = 90.000				
	γ = 90.000				

**Figure 1.** (a) Enthalpy-pressure diagram of RuN₄ with respect to the *P* $\bar{1}$ phase and (b) volume-pressure relation for RuN₄.

occupations of N atoms are 2*i* (0.859, 0.998, 0.858) and 2*i* (0.945, 0.361, 0.825). As Figure 2a, in the *P* $\bar{1}$ -RuN₄ phase at 50 GPa, each ruthenium atom is coordinated with six adjacent nitrogen atoms to form a distorted RuN₆ octahedron. In the RuN₆ octahedron, the shortest Ru–N bond length is 1.94 Å, which is smaller than the sum of the

covalent radii of Ru ($r = 1.25$ Å) and N ($r = 0.75$ Å), suggesting that there may exist a Ru–N covalent bond. From Figure 3(c), the N atoms form the extended chains (N_∞) with three different N–N bond distances of 1.32, 1.33, and 1.45 Å, which are close to those of 1.32, 1.39, and 1.43 Å in the *C*2/*c*-CsN₂ helical nitrogen chain [58] and also approach those (1.32 and 1.35 Å) in BeN₄ [21], suggesting that the N–N bond should be considered as a combination of the N–N single bond and the N = N double bond. Therefore, the *P* $\bar{1}$ -RuN₄ crystal contains the distorted RuN₆ octahedron and chain (N_∞). In the *P*4/*mbm*-RuN₄ structure at 200 GPa, the lattice constants are $a = b = 4.080$ Å and $c = 3.403$ Å. Additionally, Ru atom takes the Wyckoff 2*b* position (0.000, 0.000, 0.500) while N atoms occupy two nonequivalent sites in this structure, the Wyckoff 8*k* (0.880, 0.800, 0.500) and 8*k* (0.880, 0.620, 0.198) sites. In addition, the ruthenium atom is surrounded by eight equidistant nitrogen atoms to form the RuN₈ rectangular unit and at the centre of RuN₈ rectangular (see Figure 2b). This predicted crystal structure of the *P*4/*mbm* phase under high pressure is similar to those of the pre-existing *P*4/*mbm*-CdN₂ [29] and *P*4/*mbm*-CaN₄ [59] structures. In the RuN₈ rectangular, the equal Ru–N separation of 1.92 Å is also slightly smaller than the sum of the Ru and N covalent radii, disclosing the possible existence of a Ru–N covalent bond. Meanwhile, there is an N₂ structural unit consisting of two equivalent N atoms (similar to the diatomic molecule N₂) with an N–N bond length of 1.39 Å, which approaches that of N–N in the *cg*-N structure (1.35 Å) [8] and is between the bond lengths of N–N single bond (1.45 Å) and N–N double bond (1.25 Å), which is closest to the N–N single bond. Therefore, the *P*4/*mbm*-RuN₄ phase is composed of the RuN₈ rectangular and N₂ unit.

To investigate the effect of external pressure on the lattice parameters and volume of RuN₄, the normalised parameters a/a_0 , b/b_0 , c/c_0 , and volume V/V_0 of two

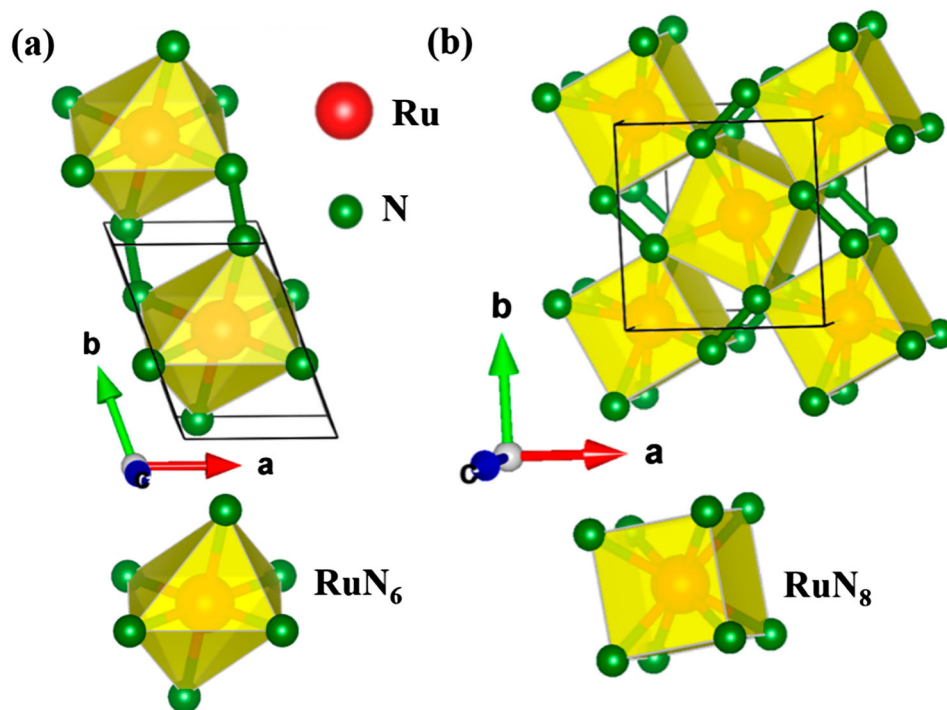


Figure 2. The crystal structure and coordination environment of two stable RuN_4 at high pressure: (a) $P\bar{1}$ - RuN_4 phase at 50 GPa and (b) $P4/mbm$ - RuN_4 phase at 200 GPa.

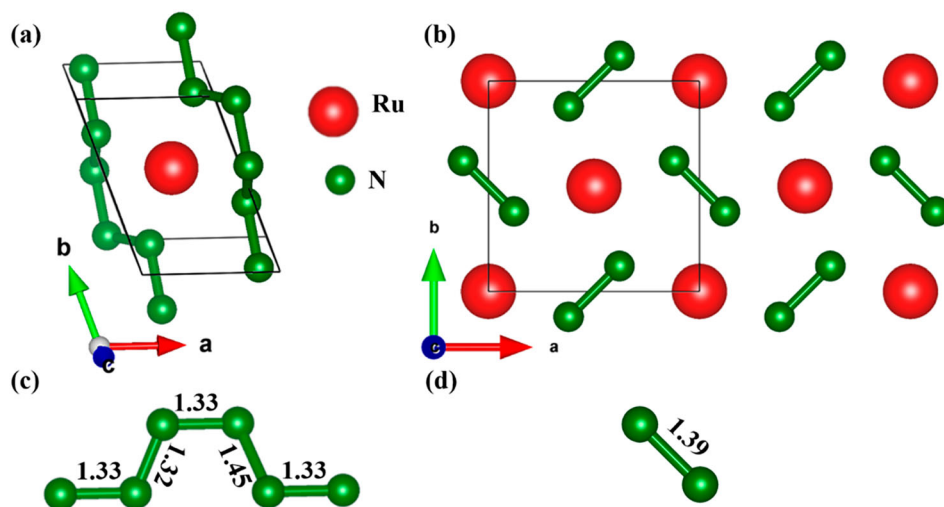


Figure 3. Crystal structures and N—N bond lengths of two types of stable RuN_4 at high pressure: (a) $P\bar{1}$ - RuN_4 crystal structure, (b) $P4/mbm$ - RuN_4 crystal structure, (c) N—N bond length of $P\bar{1}$ - RuN_4 at 50 GPa and (d) N—N bond length of $P4/mbm$ - RuN_4 at 200 GPa.

stable phases as a function of pressure are shown in Figure 4, where a_0 , b_0 , c_0 , and V_0 are the equilibrium structural parameters at 40 GPa, respectively. The results show that the a/a_0 , b/b_0 , c/c_0 , and V/V_0 values decrease gradually with increasing pressure. The primary reason for this is that the distance between the atoms decreases with increasing pressure, which increases the repulsion between the atoms and leads to an increase

in the incompressibility of the crystals under pressure. In addition, the variation processes of lattice constants for $P\bar{1}$ - RuN_4 are various while the changing trend of the lattice constant a in $P4/mbm$ - RuN_4 is close to that of the lattice constant c . Therefore, the compression on the different axes in these two stable phases should be anisotropic and the $P\bar{1}$ - RuN_4 structure have stronger anisotropy than that of the $P4/mbm$ - RuN_4 configuration.

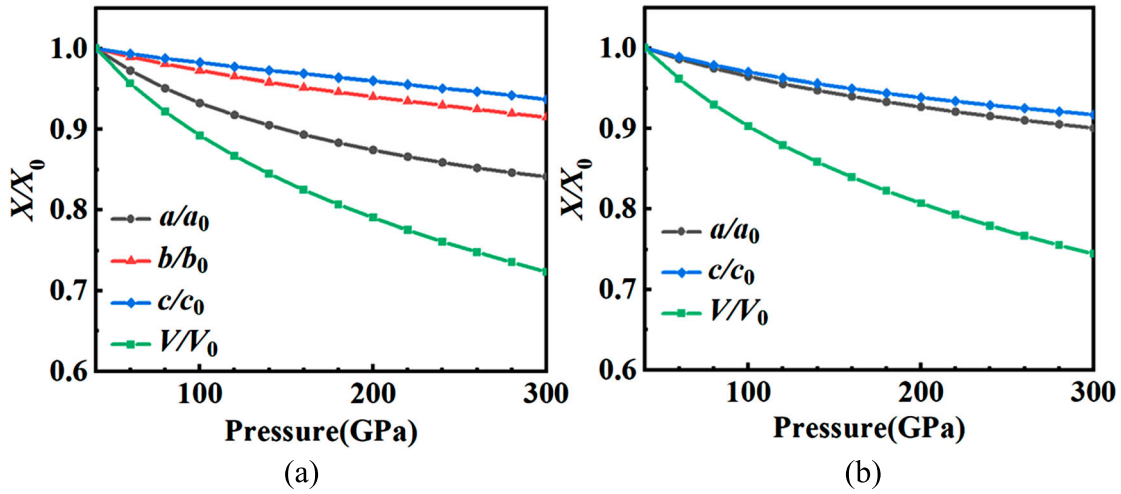


Figure 4. The normalised parameters X/X_0 ($X = a, b, c$, and V) of RuN₄ for (a) the $P\bar{1}$ -RuN₄ phase and (b) the $P4/mbm$ -RuN₄ phase as a function of pressure.

3.2. Dynamic stability and mechanical properties

To verify the dynamic stability of the $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ phases at considered pressures, we calculated their phonon spectra, as displayed in Figure 5. It is well known that the dynamic stability of one material depends on the presence or absence of imaginary frequencies in the phonon dispersion curves, which is the most common method to determine the dynamical stability of a material. The absence of any imaginary frequency in the whole Brillouin zone for these two stable structures confirms that they are both dynamically stable [60, 61]. In addition, to gain more insight into the dynamical stability of two stable RuN₄ structures, we calculated their vibrational infrared (IR) and Raman spectra, as displayed in Figure S1. The strongest peaks of the Raman wavenumbers are at about 1165 and 938 cm⁻¹ in the $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ structures, respectively, which may be caused by the stretching vibration of the intermolecular Ru – N bonds. In addition, the different peaks in the IR spectra may be due to the different vibrations of Ru and N atoms.

We calculated the elastic constants of the $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ phases at selected pressures using the strain–stress method [62, 63] and jointly verified the mechanical stability based on the mechanical stability criterion [64] of the crystalline system and the Born stability criterion [65], as shown in Table 2. The two stabilised phases belong to the triclinic and tetragonal crystal systems, respectively, and both satisfy the mechanical stability criterion and Born stability criterion of the corresponding crystal system, demonstrating their mechanical stabilities. To further evaluate the mechanical properties of the two phases under the corresponding pressures, we calculated the bulk modulus B , shear modulus G , B/G ,

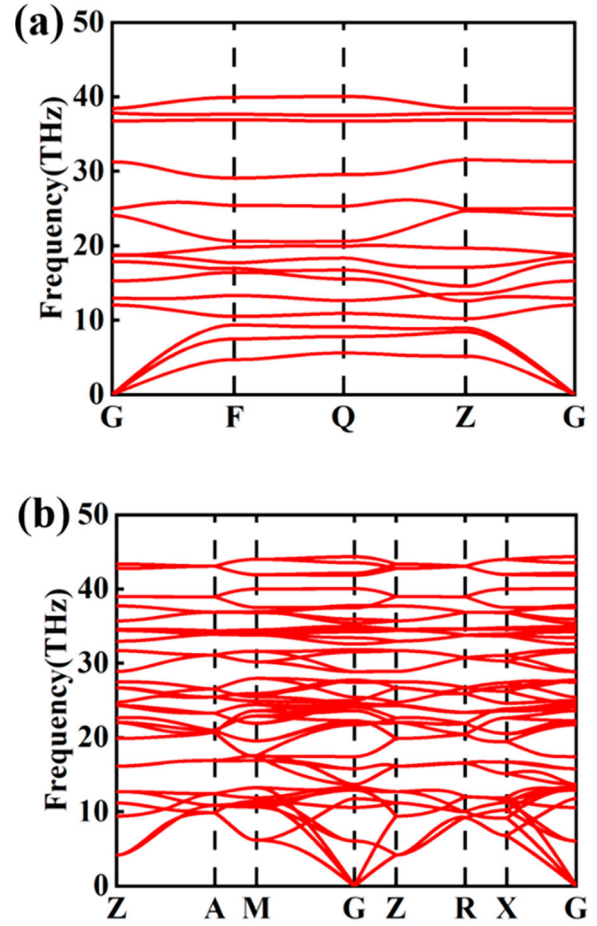


Figure 5. Phonon dispersion curves of two stable RuN₄ structures under considered pressure (a) $P\bar{1}$ -RuN₄ phase at 50 GPa and (b) $P4/mbm$ -RuN₄ phase at 200 GPa.

Young's modulus E , Poisson's ratio ν , and hardness H_v of the $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ phases (see Table 2). Generally, the values of C_{11} , C_{22} and C_{33} indicate the lin-

Table 2. Elastic constants C_{ij} (GPa), bulk modulus B (GPa), shear modulus G (GPa), Young's modulus E (GPa), B/G , Poisson's ratio ν and Vickers hardness H_v (GPa) for two stable RuN_4 structures.

Phase	$P\bar{1}$ (1 atm)	$P4/mbm$ (1 atm)	$P\bar{1}$ (50 GPa)	$P4/mbm$ (200 GPa)
C_{11}	289	538	501	1522
C_{22}	714	538	1050	1522
C_{33}	475	702	800	1625
C_{44}	275	186	442	424
C_{55}	206	186	27	424
C_{66}	173	243	200	547
C_{12}	155	149	214	446
C_{13}	205	82	336	559
C_{23}	169	82	376	559
B	255	267	434	865
G	154	217	221	483
E	385	512	567	1221
B/G	1.65	1.23	1.96	1.79
ν	0.25	0.18	0.28	0.26
H_v	18.11	33.5	18.36	34.60

ear incompressibility along the corresponding a-, b- and c-axis. Note that the C_{22} value of the $P\bar{1}$ - RuN_4 phase is higher than that of C_{11} and C_{33} , indicating that the structure has stronger incompressibility along the b-axis than along the a-axis and c-axis. Similarly, the value of C_{33} in $P4/mbm$ - RuN_4 is higher than the values of C_{11} and C_{22} , indicating that this configuration is more incompressible along the c-axis than along the a- and b-axis. C_{44} is an important indicator of the indentation hardness of crystals. The C_{44} values for both structures are greater than 400 GPa at the corresponding pressures, indicating that they should have high hardness. At ambient pressure, The predicted bulk modulus values of the $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 phases are up to 255 and 267 GPa, respectively, which are higher than those of cubic-SiC (225 GPa) [66] and close to that of $\text{Pd}_2\text{Mo}_3\text{N}$ (300 GPa) [67], indicating that these two crystal structures possess a good compressive resistance.

The shear modulus of $P\bar{1}$ - RuN_4 is 221 GPa, which is higher than that of ReN_4 (213 GPa) [68], and the shear modulus of $P4/mbm$ - RuN_4 reaches 483 GPa, implying that these two structures may be hard materials. At zero pressure, the Young's modulus values of $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 are 385 and 512 GPa, respectively, which are greater than $\text{I}43d$ - Hf_3N_4 (313 GPa) [35] and orth- CrN (378 GPa) [69], indicating that these two crystals are very rigid. We can evaluate whether the material is ductile or brittle by the B/G value. Usually, if $B/G > 1.75$, the material is ductile, otherwise it is brittle. Under atmospheric pressure, the $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 phases have a B/G of 1.65 and 1.23, respectively, higher than 1.75, illustrating that both structures are brittle [70]. In addition, the empirical Vickers hardness formula [52] is adopted to compute the Vickers hardness of these two RuN_4 stable structures (see Table 2). At ambient pressure, the hardness value of the $P\bar{1}$ - RuN_4

phase (18.11 GPa) is close to that of NaCl-type HfN (19.0 GPa) [35] and higher than that of GaN (15.10 GPa) [71]. More importantly, the hardness value of $P4/mbm$ - RuN_4 at atmospheric pressure is 33.50 GPa, which is much higher than that of TcN_4 (21.20 GPa) [72], and close to those of BP (31.2 GPa) [73], YB_{12} (33.16 GPa) [74] and $\text{C2}/m$ - B_{13}N_2 (37.3 GPa) [75]. The consequences above uncover that two stable structures are promising hard materials [65].

Elastic isotropy is an important physical property of a material. To reflect the arrangement of atoms along the corresponding direction and quantify the degree of isotropy, we calculated the general elastic anisotropy index of crystals with arbitrary symmetry. Generally speaking, for isotropic solids, their A^U is zero. The degree of anisotropy is expressed by the deviation from zero. The greater deviation from zero corresponds to the higher degree of anisotropy of the material. In addition, the shear anisotropy constant can reflect the anisotropy of a solid in different planes. For isotropic crystals, the values of A_1 , A_2 , and A_3 are equal to 1, and for any crystals less than or greater than 1, the degree of shear anisotropy of the crystal is indicated [76]. Both the anisotropy index and the shear anisotropy factor are listed in Table S1. The consequences indicate that the anisotropy of $P\bar{1}$ - RuN_4 is relatively more significant, while $P4/mbm$ - RuN_4 exhibits relatively lower degree of anisotropy, which is agreement with the result about the influence of external pressure on the lattice parameters.

3.3. Electronic properties and bonding characteristics

In order to elucidate the electronic properties of the $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 structures, we probed into their electronic band structures and density of state (DOS), as plotted in Figure 6. In the band structures of both phases, the bottom of the conduction band and the top of the valence band overlap across the Fermi level, indicating they are metallic. This conclusion is also demonstrated by the DOS, as there exists a finite total density of states at the Fermi level. The DOS near the Fermi level is mainly contributed by the electrons in the Ru-4d and N-2p orbitals. Meanwhile, the curve of Ru-4d state is similar to that of the N-2p state in the whole energy region for these two stable structures, which suggests there exists a strong between these two orbitals. This phenomenon uncovers a strong covalent interaction between Ru and N atoms, which is in accord with the analysis of the Ru - N bond length.

The electron localization function (ELF) is an effective method for analysing the interatomic bonding feature. Hence, we computed the ELFs for the $P\bar{1}$ - RuN_4

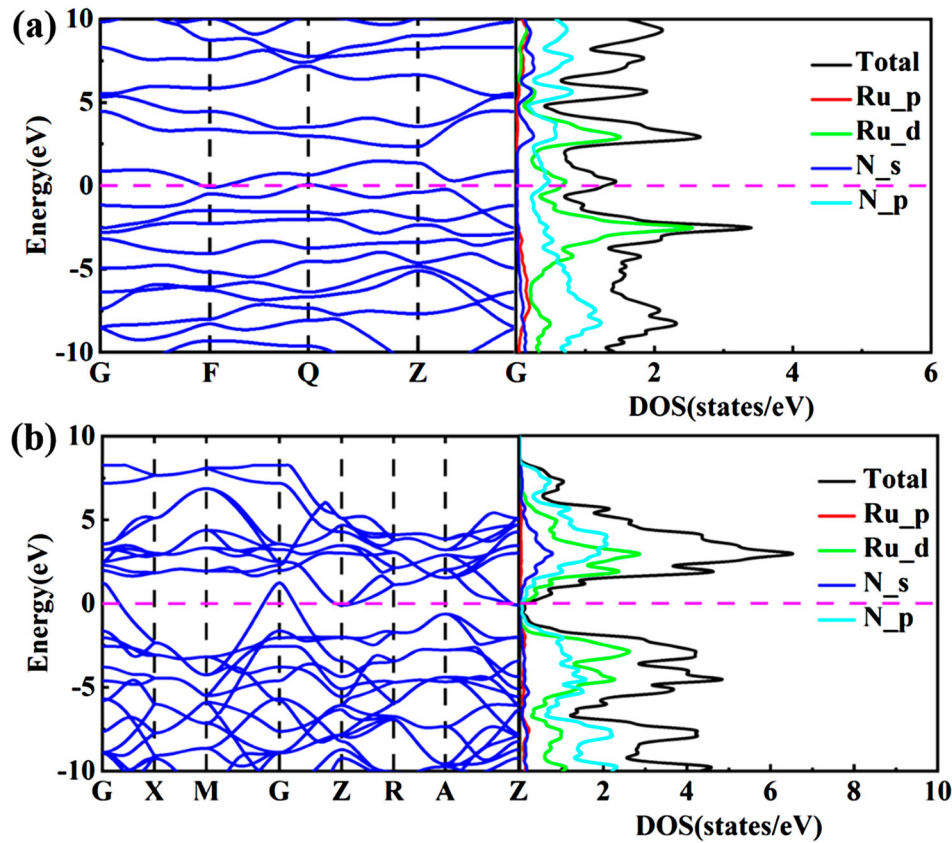


Figure 6. The band structure and projected density of states of two stable RuN_4 structures at high pressure (a) $P\bar{1}$ - RuN_4 phase at 50 GPa and (b) $P4/mbm$ - RuN_4 phase at 200 GPa. The point, where the energy value is zero, is the Fermi level.

and $P4/mbm$ - RuN_4 structures to clarify the Ru–N and N–N bond characters, as depicted in Figure 7. It is seen that high electron localisation is in the region close to the N atom between Ru and N atoms. At the same time, similar electron localisation exists in the region between N atoms. These two consequences above reveal the strong Ru–N and N–N bonds in two considered RuN_4 phases. The Ru–N and N–N covalent bonds in two stable RuN_4 structures are responsible for the high hardness.

The crystal orbital Hamilton population is an effective means to probe into the bonding, antibonding, and non-bonding interactions of materials. To further substantiate the covalent bonding characteristics in the $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 structures, we calculated their integrated-crystal orbital Hamilton populations (ICOHP) of the Ru–N and N–N bond at the Fermi level, as shown in Figure 8. On the one hand, the ICOHP values of the N–N pair in $P\bar{1}$ - RuN_4 (−14.28 eV) and $P4/mbm$ - RuN_4 (−11.70 eV) are close to those of the N–N pair in M' - ReN_8 (−14.20 eV) [30] and $Immm$ - MnN_4 (−12.024 eV) [77]. On the other hand, the ICOHP values of the Ru–N pairs in $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 are −3.90 eV and −3.53 eV, respectively, which are lower than those of the

Re–N pair in M - ReN_8 [30] (−3.0 eV). These two consequences illuminate that the $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 phases consist of the Ru–N and N–N covalent interactions and the N–N covalent bond is stronger than the Ru–N bond. Moreover, the high elastic modulus and Vickers hardness may be associated with the existence of the Ru–N and N–N covalent bonds.

3.4. High energy density

Polymeric nitrogen are usually considered to be HEDMs and they could decompose at ambient conditions with the exothermic process because of the energy difference between the single N–N bond and triple $\text{N}\equiv\text{N}$ bond. Based on the high concentration and existence of N–N single bonds in these two RuN_4 structures, they are possible HEDMs. In the reaction process of $2\text{RuN}_4 \rightarrow 2\text{Ru} + 3\text{N}_2$ (see Table 3), the calculated energy densities of $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 can reach 1.71 and 2.81 kJ/g, respectively. The former is slightly lower than that of FeN_6 (1.83 kJ/g) [78], but higher than that of BaN_4 (0.626 kJ/g) [79] and the latter is comparable to that of LiN_5 (2.72 kJ/g) [80], and higher than those of $P\bar{1}$ - MgN_4 (2.08 kJ/g) [81] and $P\bar{1}$ - CoN_4 (2.36 kJ/g) [34].

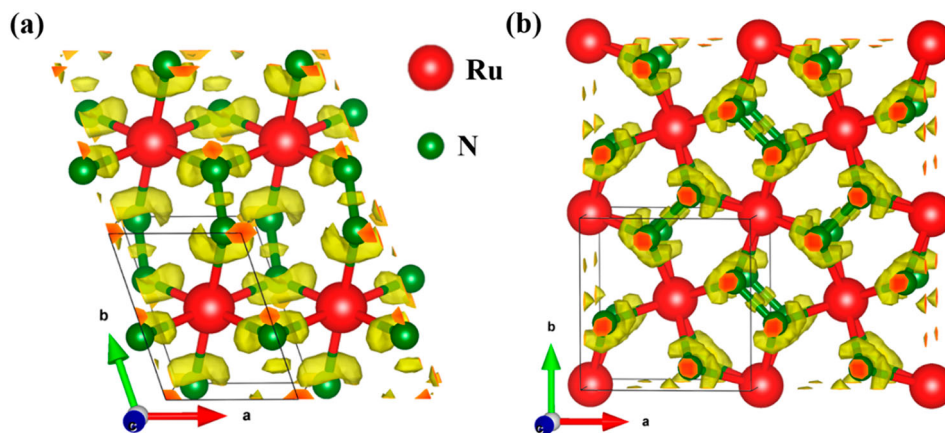


Figure 7. Electron localisation functions for two stable RuN_4 structures at high pressure. The isosurface value is 0.80. (a) $P\bar{1}$ phase at 50 GPa and (b) $P4/mbm$ phase at 200 GPa.

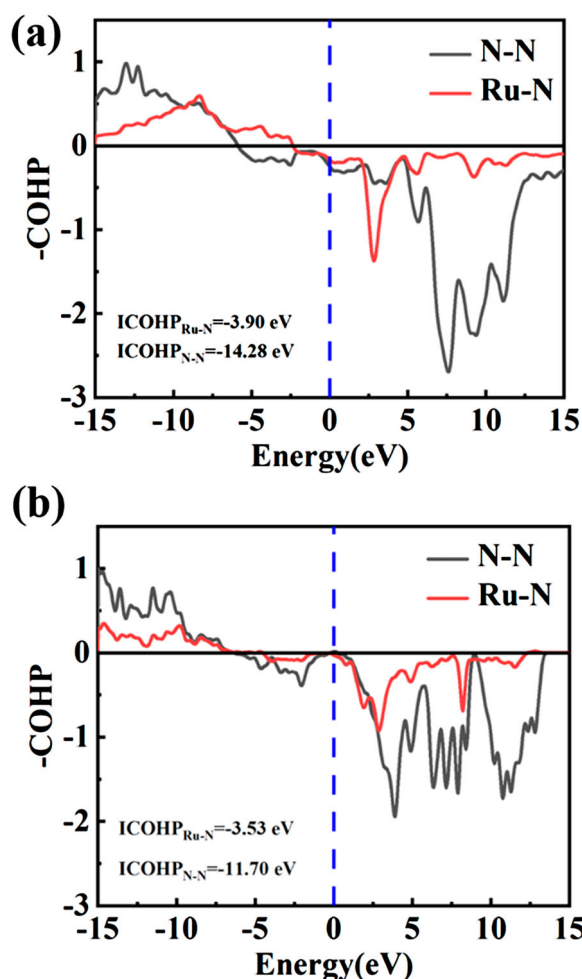


Figure 8. Crystal orbital Hamiltonian population (COHP) analysis of two stable RuN_4 structures at high pressure (a) $P\bar{1}$ - RuN_4 at 50 GPa and (b) $P4/mbm$ - RuN_4 at 200 GPa. The point, where the energy value is zero, is the Fermi level.

In addition, we further the volumetric energy density for these two RuN_4 phases, as tabulated in Table 3, in conjunction with the results of the famous explosive. The

Table 3. Comparative analysis of energy density and volumetric energy density of various polymeric nitrogen structures.

Polymeric nitrogen	Energy density (kJ/g)	Volumetric energy density (kJ/cm ³)
$P\bar{1}$ - RuN_4	1.71	12.20
$P4/mbm$ - RuN_4	2.81	18.70
FeN_6 [78]	1.83	
C2/c-BaN_4 [79]	0.626	
LiN_5 [80]	2.72	
$P\bar{1}$ - MgN_4 [81]	2.08	
$P\bar{1}$ - CoN_4 [34]	2.36	
$P\bar{3}1c$ - CoN_8 [34]	6.14	
cg-N [6,82]	9.70	
TNT [83]	4.30	7.20–8.00
HMX [83]	5.70	10.83
RDX [84,85]		10.10
PETN [84,85]		10.60

estimated volumetric energy densities for the $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 configurations are 12.20 and 18.70 kJ/cm³, which are higher than those of TNT (7.2–8.0 kJ/cm³), cyclotrimethylene trinitramine (RDX) (10.10 kJ/cm³) and pentaerythritol tetranitrate (PETN) (10.60 kJ/cm³) [84, 85] values, revealing its high-energy density characters. Hence, the high-energy density property of $P4/mbm$ - RuN_4 is superior to that of $P\bar{1}$ - RuN_4 .

4. Conclusions

In this work, we successfully predicted two new RuN_4 phases, $P\bar{1}$ - RuN_4 and $P4/mbm$ - RuN_4 based on the USPEX method combined with first-principles calculations, which enriches the high-pressure phase of the Ru–N system. Within the pressure range of 0–300 GPa, the stable $P\bar{1}$ - RuN_4 phase does not appear until 37 GPa and subsequently transforms into the $P4/mbm$ - RuN_4 configuration when the pressure reaches 162 GPa. Interestingly, the $P\bar{1}$ - RuN_4 structure contains the RuN_6 distorted octahedra and extended chains (N_∞) while the $P4/mbm$ - RuN_4 phase consists of the RuN_8

rectangles and the N₂ units. Calculations of elastic constants and phonon spectra confirm that the $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ structures are mechanically and dynamically stable. The hardness value of the $P4/mbm$ -RuN₄ phase at atmospheric pressure is up to 33.50 GPa. Electronic structure calculations demonstrate that both the $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ phases are metallic. The analyses of electron localisation function and crystal orbital Hamiltonian abutment analyses display that both $P\bar{1}$ -RuN₄ and $P4/mbm$ -RuN₄ phases comprise a combination of the covalent Ru–N and N–N bonds. On decomposition, $P4/mbm$ -RuN₄ is expected to release a significant amount of energy (2.81 kJ/g) with a volumetric energy density of 18.70 kJ/cm³. Hence, $P4/mbm$ -RuN₄ is a new multifunctional material that has both high hardness and high energy density.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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